

**Épreuve du 1^{er} groupe****ANGLAIS****Automatic Milking Systems**

A robotic milking system also known as an Automatic Milking System (AMS) allows cows to choose when to be milked by a robotic system, without the direct assistance of milking staff. It has a robotic arm and an identification system. An AMS needs good farm management to be successful and works best when motivated by cow appetite. For your farm, consider phone and internet coverage, the landscape, proximity to servicing, and if you're tech-friendly. AMS set up requires planning farm layout, water access, dairy design, and smart pasturage management.

When a cow enters the milking stall, she is identified via a tag strapped to her leg, around her neck or in her ear and checked whether she's due for milking. If the software determines that the cow has milking permission, her teats are cleaned and the teat cups attached, one at a time. The machine milks her, collects data on both the cow and milk quality, then sends the milk to separate tanks, if requested by the operator using the software settings. While milking, the cow may be offered a food concentrate (meal, grain, molasses) as a reward for willing to be milked. If this is not yet her milking time, the front gates open and the cow is released from the stall.

Automatic milking relies on cows "volunteering" for milking in the AMS and on distributing their milking appointments throughout 24 hours. The main motivator for cow movement around the farm is feed: either fresh pasture and/or food concentrate offered in the AMS. Most AMS farms are set up with two or three new areas of pasture available to the herd daily. Cows access pasture after visiting the AMS or can be diverted directly to the new pasture if milking permission is not granted. So, a small level of concentrate is generally necessary to maintain cow circulation through the milking stations, although there is one farm in New Zealand operating successfully with only grass and no feed in the AMS.

Adapted from <https://www.dairynz.co.nz/milking/new-dairy-technology/automatic-milking-systems/>

I. COMPREHENSION (10 marks)

A) Find information in Paragraph 1 to show that the statements below are all TRUE.

3 marks

1. The time when the system starts the automatic milking process is decided by an animal.

☞ _____

2. The system does not need human intervention.

☞ _____

3. It is recommended that farmers know develop familiarity with technology before using AMS.

☞ _____

B) Read Paragraph 1 again and find equivalents of the following French expressions.

2 marks

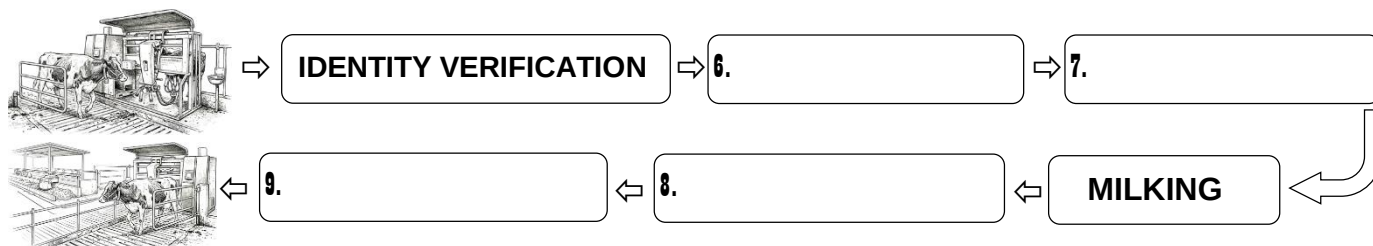
4. « exploitation agricole » : ☞ _____

5. « voisinage avec des services de maintenance » : ☞ _____

C) Read Paragraph 2 and fill in this flow chart with appropriate phrases from the box.

2 marks

Milking • Tanking Phase • Data Collection • Permission Check • Identity Verification • Hygiene Phase



D) Each of the actions listed in Column Y is dependent on another action mentioned in Paragraph 3. Find that missing information and fill it in Column X, as in the example.

3 marks

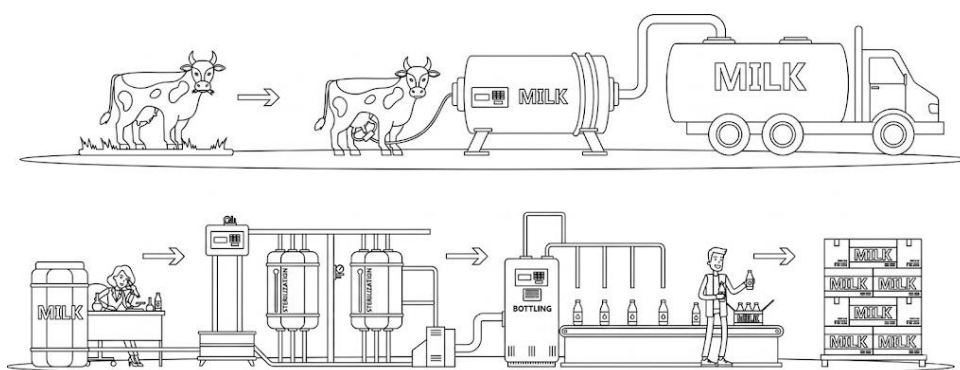
Column X – Chief Actions		Column Y – Dependent Actions
<i>Example:</i> Cow Volunteerism	☞	Automatic Milking
Feed	☞	10.
11.	☞	Access to Pasture
12.	☞	Refusal of Milking Permission
Availability of concentrate	☞	13.

II. LINGUISTIC and COMMUNICATIVE COMPETENCE (06 marks)

E) Look at this illustration of the milk production process and complete its commentary with appropriate phrases from the box.

2 marks

for transport • bottling line • automated machinery • milk production • laboratory testing



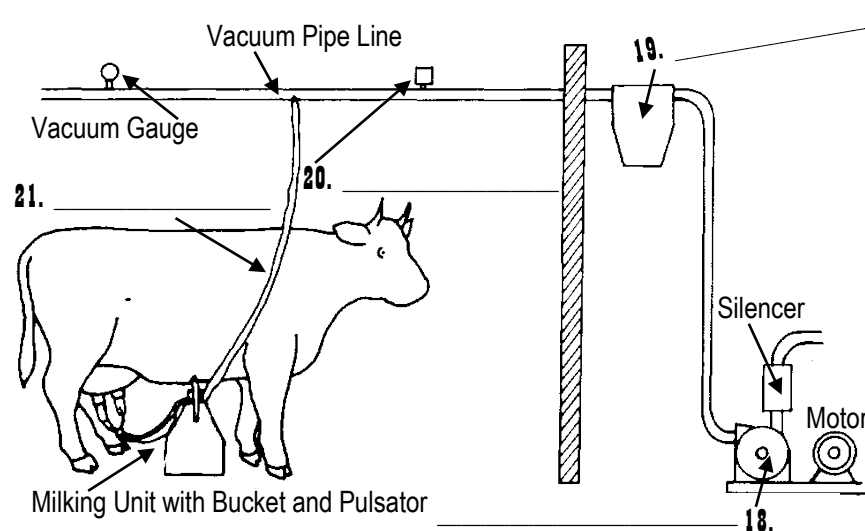
The process chart illustrated in the image summarizes the step-by-step line of milk production, from the farm to the final packaged product. The chain begins with cows

eating grass in a pasture, which are then brought in stalls for the milking stage using (14) _____. The collected milk is transferred to a large storage tank before being loaded into a specialized milk truck (15) _____. Upon arriving at the processing facility, the milk undergoes strict (16) _____ and quality control. Once approved, it passes through a sterilization system to confirm it is safe for consumption. Finally, the milk moves to an automated (17) _____, where it is packaged into individual bottles, boxed into crates, and stacked onto pallets for distribution.

.../...3

F) The image below is a representation of how mechanical milking systems work. The descriptive text lists the different parts of the system and explains the role of each part. Complete the legend of the image appropriately with the bold words in the text.

2 marks



Based on the diagram, the mechanical milking system relies on a continuous vacuum flow generated between the engine room and the milking parlour to safely extract and collect milk. Powering the entire operation, the **motor** drives the **pump**, which creates the essential negative pressure required for the system to work. Any operational noise from the pump is muted by the **silencer**.

The vacuum is pulled through the overhead **vacuum pipe line**, which is safely partitioned from liquid entry by a **moisture trap** designed to catch any accidental condensation or fluids. To guarantee safety and efficiency during operation, the **vacuum gauge** shows a visual readout of the pressure level, while the **vacuum regulator** automatically adjusts to maintain a stable, safe pressure. This regulated suction travels through the flexible **vacuum tube** directly to the **milking unit with bucket and pulsator**. Here, the teat cups attach to the cow's udder, the pulsator mimics a natural suckling rhythm by alternating pressure, and the bucket safely collects the raw milk.

G) Complete this passage with the appropriate options from those proposed in parentheses.

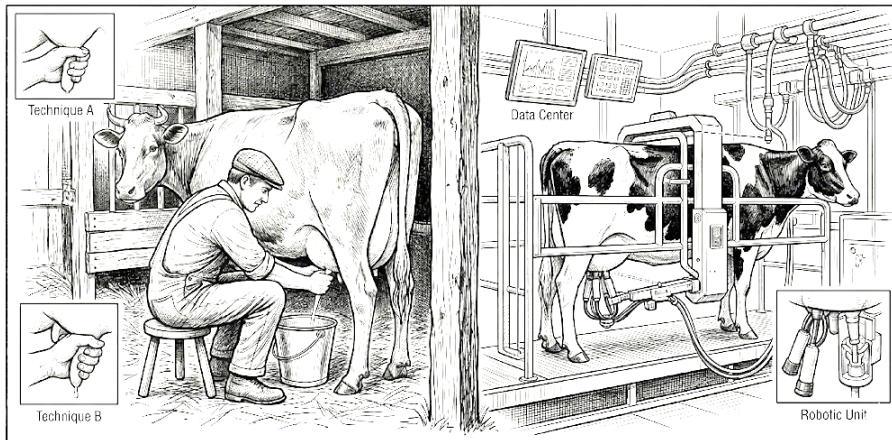
2 marks

Automated Milking Systems (AMS) are the result of the implementation of advanced technology related to robotics to increase milk production through improved efficiency and automation. These technologies are developed (22) _____ (in ☼ as ☼ for) response to the growing market opportunities for the worldwide milk industry, which is projected to grow by 35% by 2030. However, global demands will also be (23) _____ (accompanying ☼ accompanied ☼ accompany) by 14 million traditional milk farms shutting down production due to high competitiveness and requirements for guaranteed milk quality and animal well-being. Such well-being is considered a very serious concern for consumers, (24) _____ (what ☼ which ☼ that) is accomplished by AMS since it is based on the "milking when they like" system that protects and improves the well-being of cows. (25) _____ (For example, ☼ Because, ☼ Though,) modeling heat stress in AMS has concentrated recently on the rumination and milking performance, identifying production factors and thermal comfort indices.

Adapted from www.pmc.ncbi.nlm.gov/articles/PMC7285505/

III. WRITING (04 marks)

Choose ONE topic and write between 100 and 150 words about it.



Topic 1: Some animal protection activists protest against automatic milking systems because, they say, AMS machines help farmers abuse and over-exploit animals. They compare the condition of these animals with that of Black people during slavery. Do you agree with these objections? Why or why not?

Topic 2: Take a look at this image and write a comparative essay on the similarities and differences between traditional milking and modern automated milking systems.

[illegible]